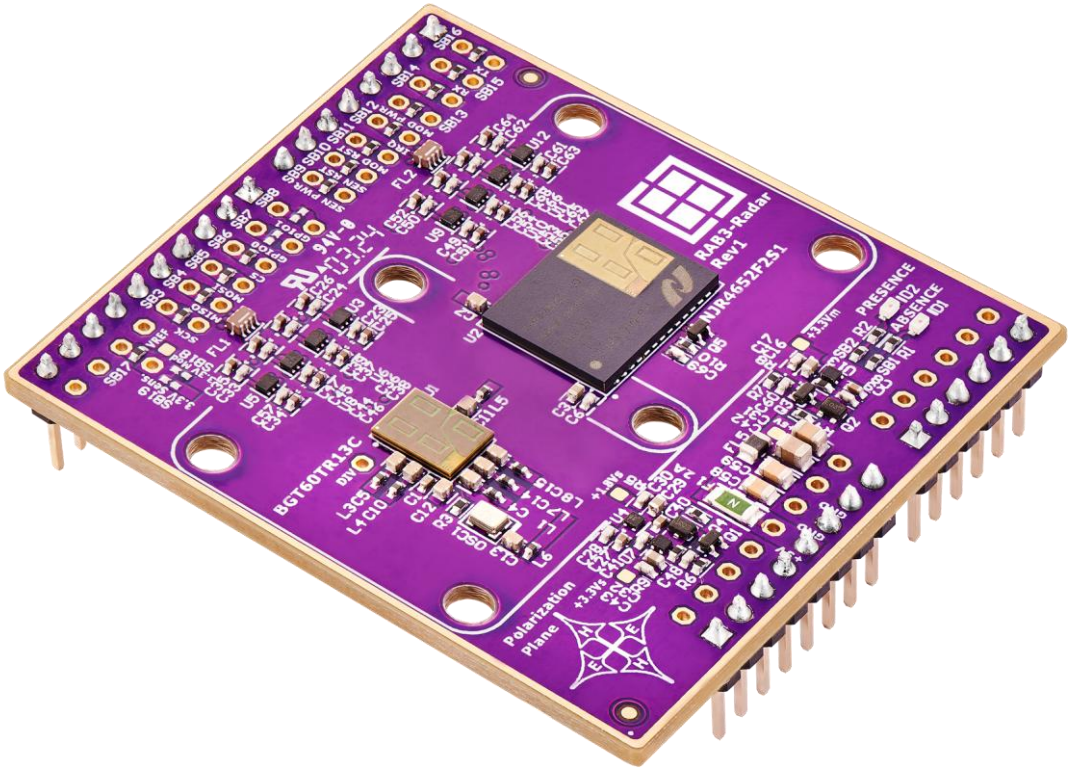
Water level measurement with radar sensor

Application note



Versions

Version	Date	Rationale
1.0	April 16, 2026	First release. Author: ROJ, KOA
1.1	October 22, 2026	Short range level measurement. Author: ROJ, KOA

Legal disclaimer

The evaluation board is for testing purposes only and, because it has limited functions and limited resilience, is not suitable for permanent use under real conditions. If the evaluation board is nevertheless used under real conditions, this is done at one's responsibility; any liability of Rutronik is insofar excluded.

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Introduction

This document describes the calculations behind the water level measurement in the tank using Infineon BGT60TR13C radar sensor.

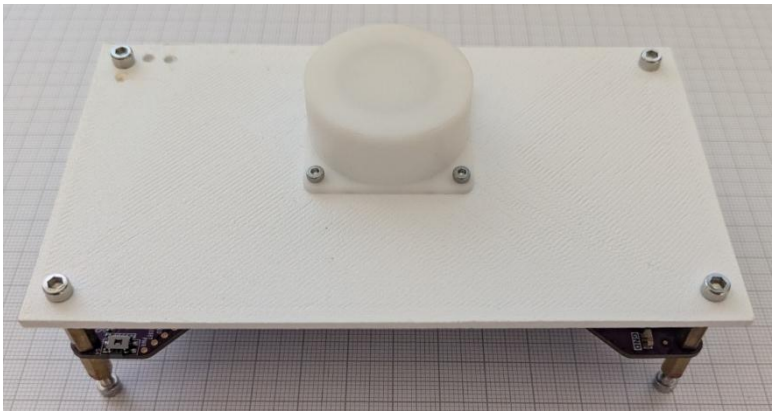
Useful links

1. GitHub project with the source code (firmware+GUI): https://github.com/RutronikSystemSolutions/RDK2_RAB3_Radar_Level_Measurement
2. Application note to understand the basics of digital signal processing for FMCW radar, including step-by-step explanation of turning frequency into range: https://github.com/RutronikSystemSolutions/RAB3-Radar_Documents/blob/main/RAB3-Radar_AppNote_Signal_Processing.pdf

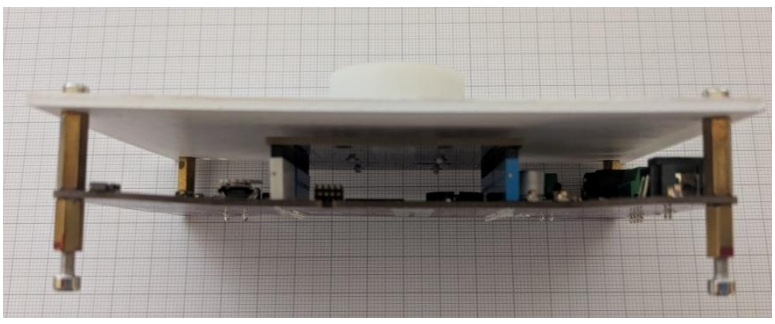
Hardware

The demonstrator is made of a RDK2 and RAB3-Radar boards.

To focus the field of view, a +/-6° lens has been used.

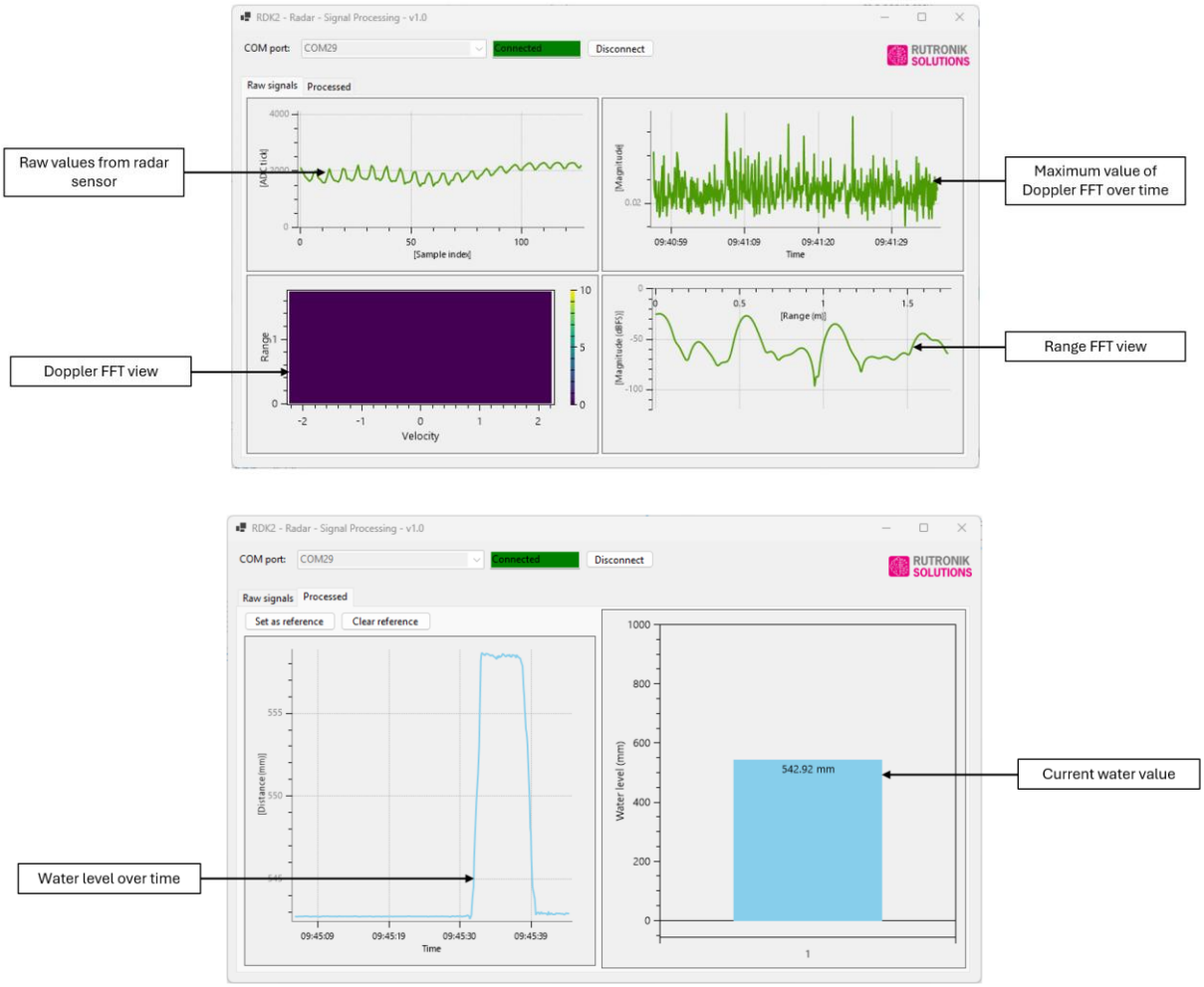


Lens enabling to focus the field of view to +/-6°



RDK2 and RAB3-Radar boards

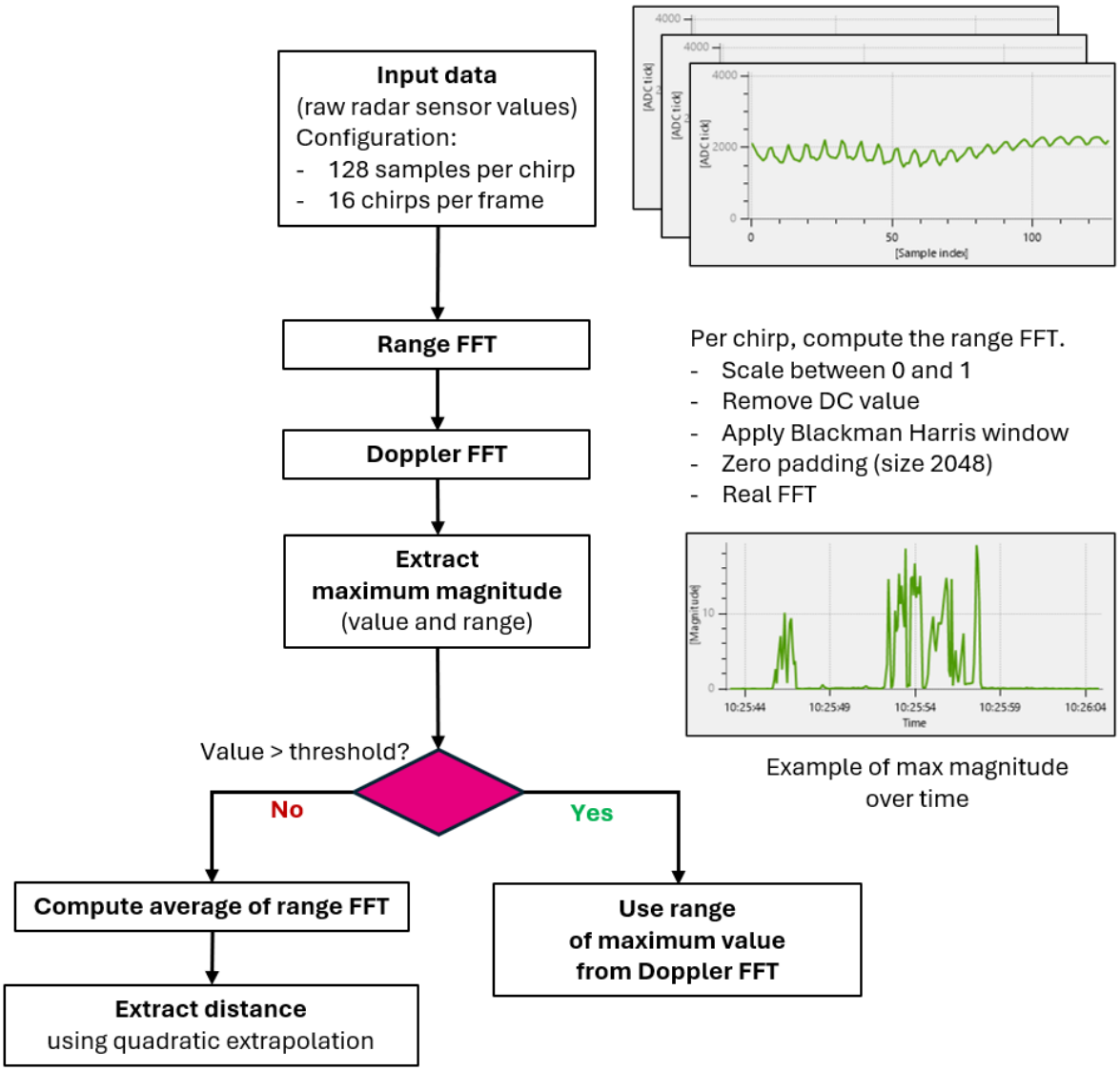
GUI overview



Extract the water level from the raw data

We do not directly extract the level of water from the raw sensor data, but we compute the distance from the radar sensor to the water.

The algorithm is shown below.



No motion case

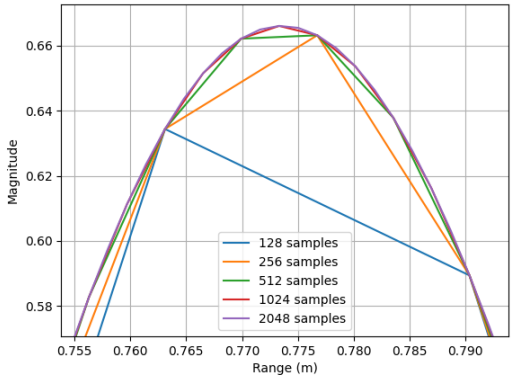
In case of no motion, we use the information of the range FFT to compute the distance. This is the more precise way.

We first compute the range FFT for all chirps using zero-padding. Next, we calculate the average value (complex values!) and then the frequency that has the more reflected signal using quadratic extrapolation.

Why using zero-padding

Our goal is to extract the distance. In radar processing world, the distance can be computed using the beat frequency of the reflected signal. The higher the resolution on the frequency analysis is, the better the resolution on the distance will be.

When using FFT to analyse frequencies inside a signal, using zero-padding enables to enhance the resolution on the frequency (drawback: more memory and computing power are necessary).

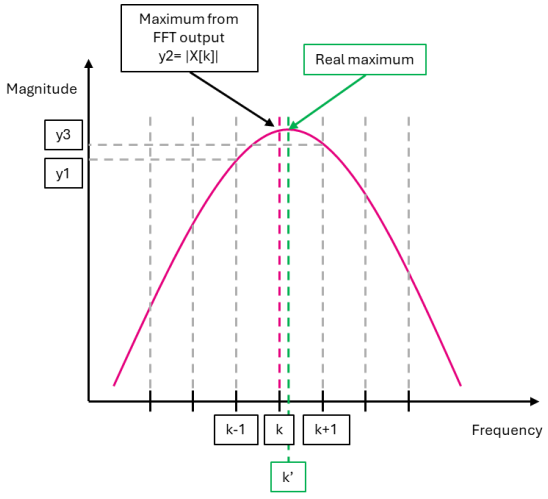


Magnitude of the FFT when using:

- 128 samples (no zero-padding)
- 128 samples + 128 zeros
- 128 samples + 384 zeros
- 128 samples + 896 zeros
- 128 samples + 1920 zeros

Quadratic extrapolation method

To compute the range, given the finite FFT output, we can estimate it using the quadratic method:



y1 = |X[k-1]|

y2 = |X[k]|

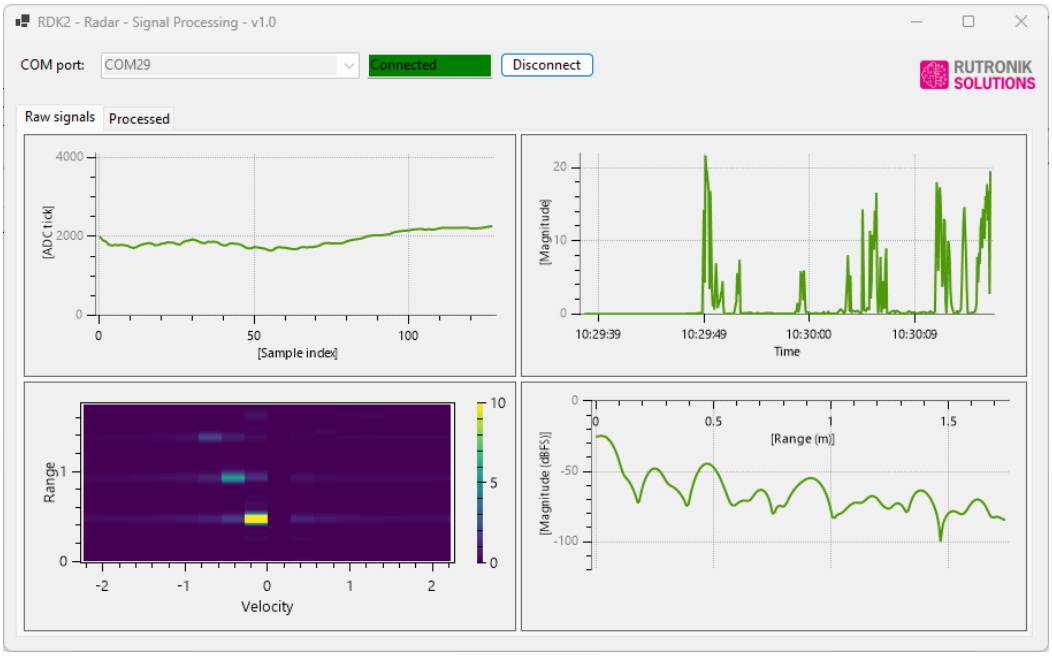
y3 = |X[k+1]|

$$d = \frac{y3 - y1}{2 * (2 * y2 - y1 - y3)}$$

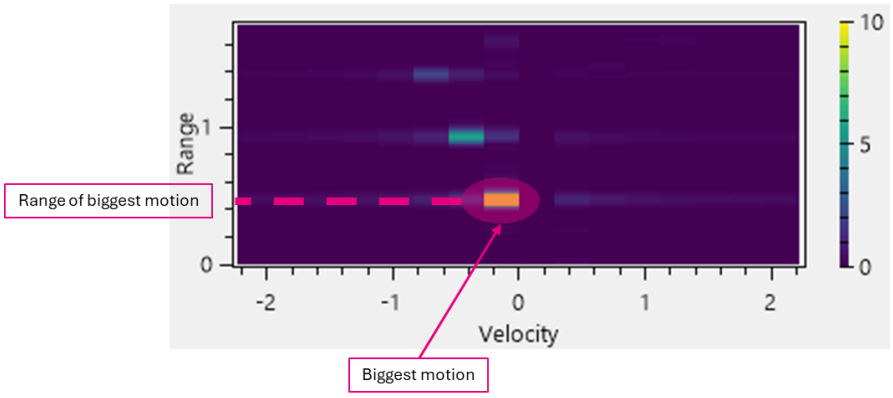
k' = k + d

Motion case

In case of motion, the range FFT is too unstable and cannot be used to obtain accurate distance measurement. Instead of using the range FFT information, we will use the Doppler FFT results, to extract the range with the biggest measured motion.



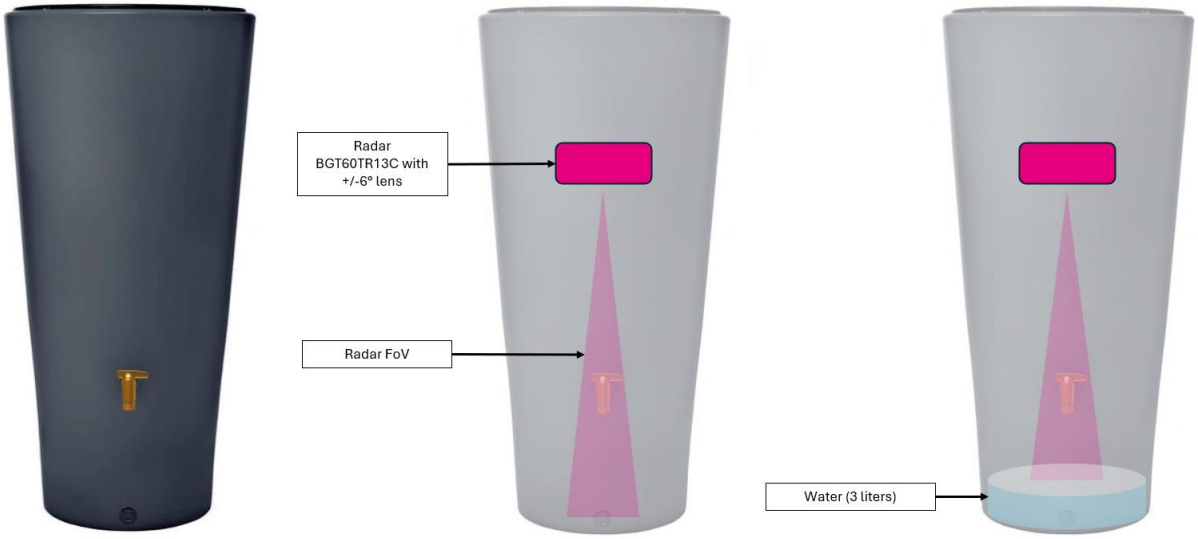
This is less accurate as the range FFT method.



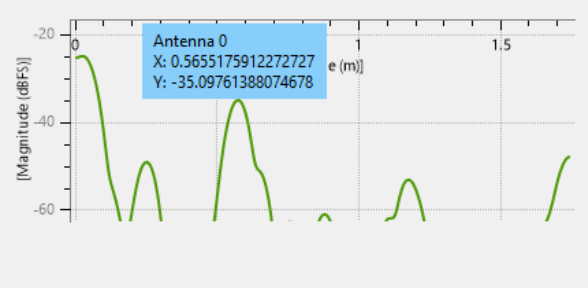
Water effect

The main questions are: does water reflects the radar wave? And if yes, how good is the reflection?

We placed a radar sensor inside a 220L water container. The distance from radar to the bottom of the container is around 550mm. We then added 3 liters of water inside the container, to see how the signal changes.

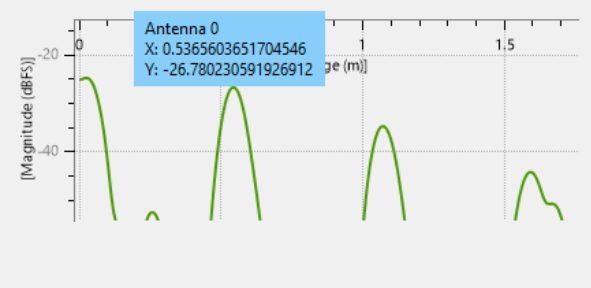


Looking at the range FFT which is used to extract the distance from radar to water, we observe:



Without water

Max magnitude = -35.1dBFS at 565 mm distance



With 3 liters of water

Max magnitude = -26.8dBFS at 536 mm distance

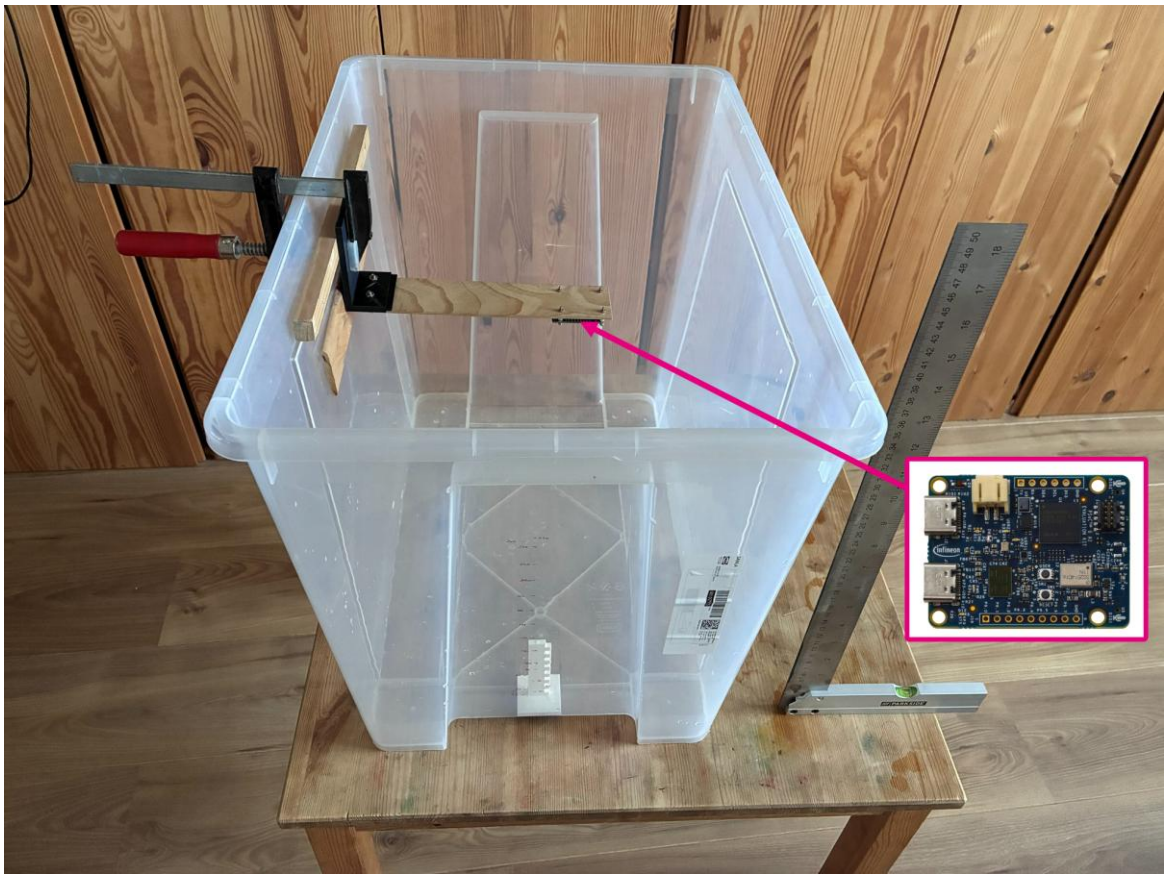
The signal reflected by water is 8.3dBFS bigger (which is more than 2 times), that means the water reflects 60GHz radar waves.

Short range level measurement

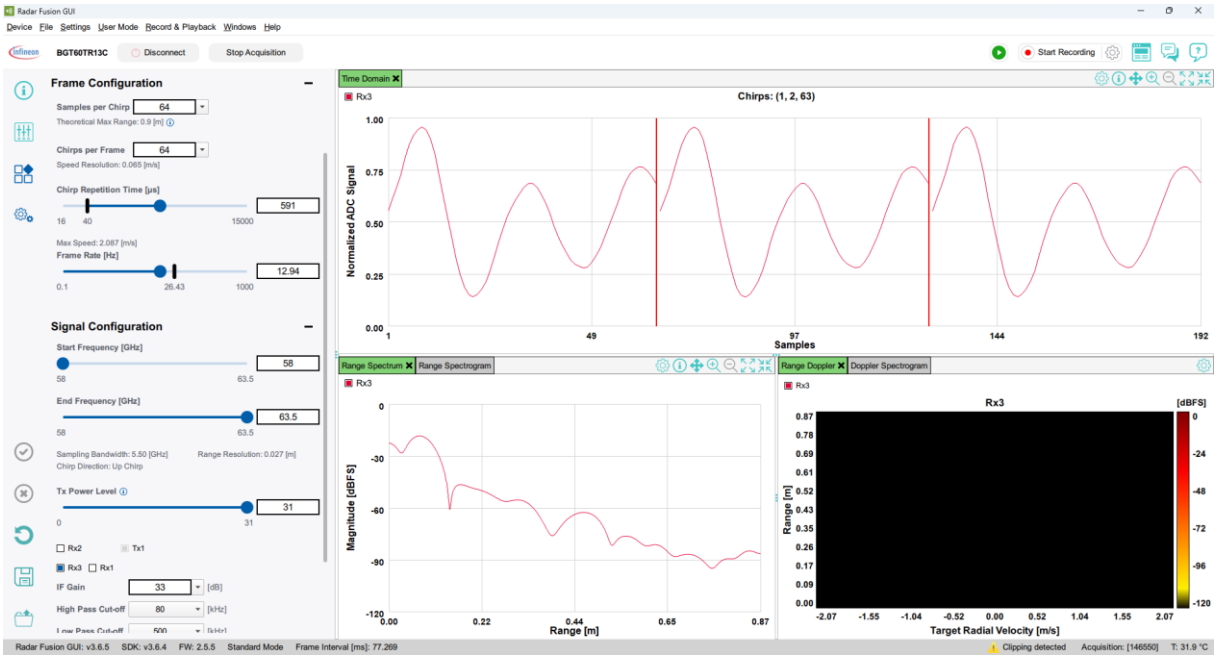
One of our tasks was to determine the minimum distance at which the radar must be positioned relative to the water's surface to ensure accurate measurement results.

If the radar is positioned too close to the water, the reflected signal may reach saturation, the ADC amplitude will approach the maximum permissible value and signal peaks will be clipped, which will reduce the quality of the measurements.

To determine the minimum distance from the radar to the water surface, we used the Infineon CY8CKIT-062S2-A development kit with the BGT60TR13C radar sensor.

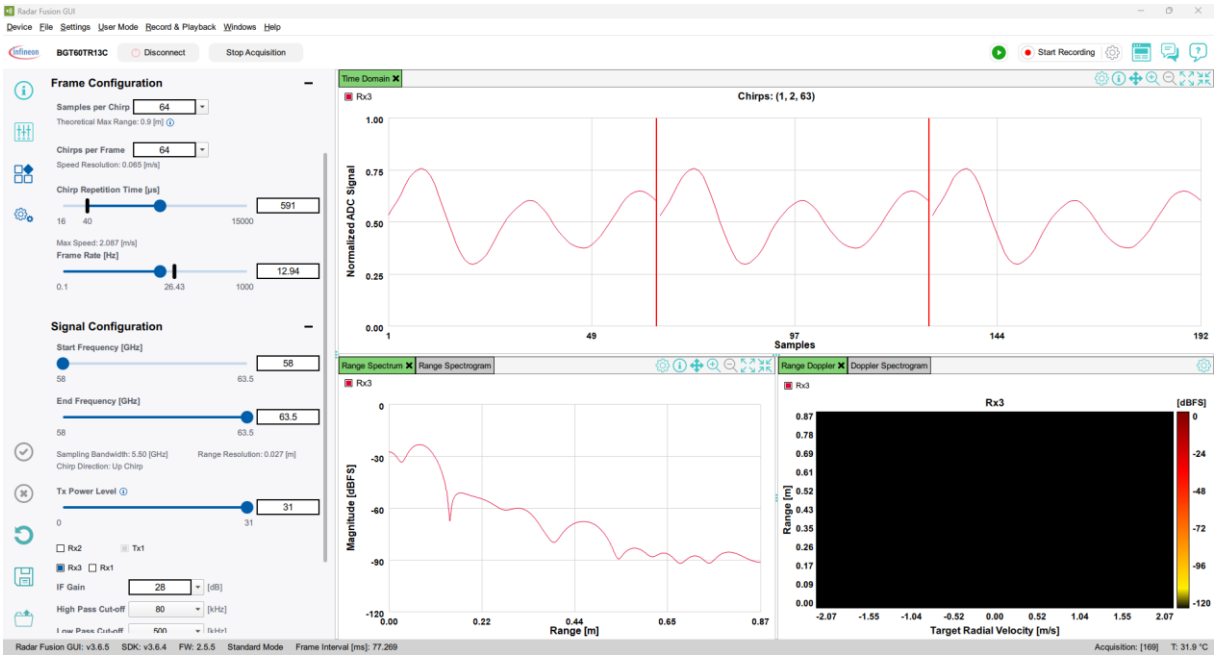


During the tests, the radar was positioned 290 mm from the bottom of the container; we filled the container with water to various levels and analysed the results in the Radar Fusion GUI. The test results are shown below.



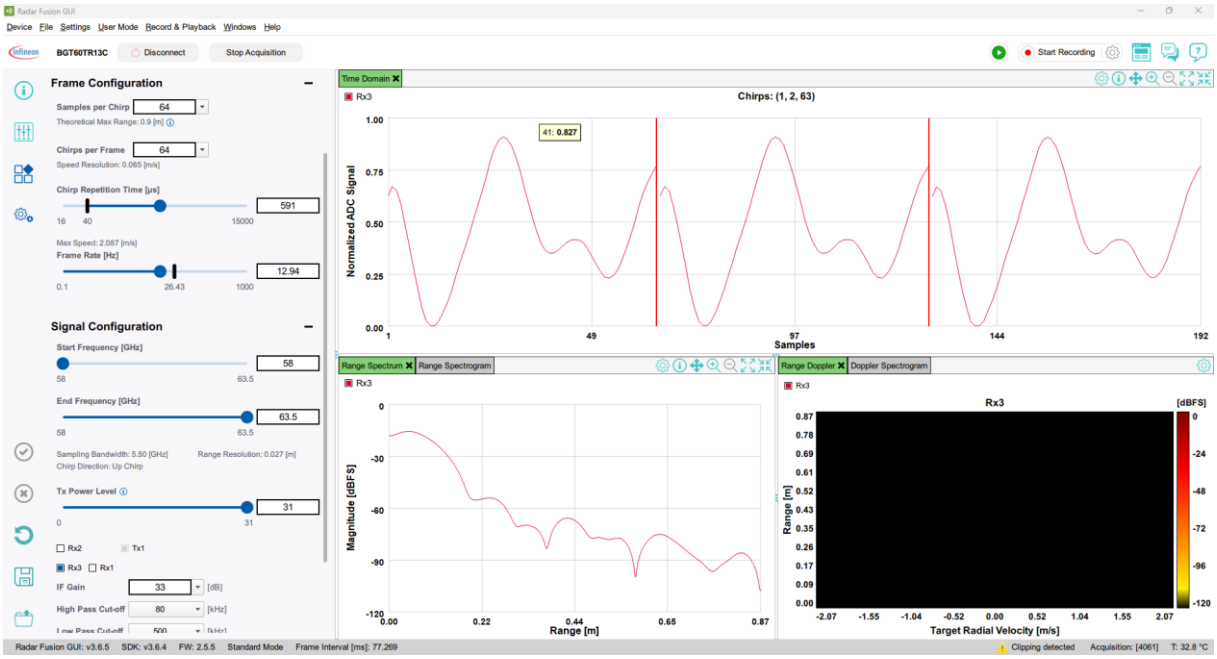
Water level = 220mm, gain = 33dB.

Result: almost saturated, clipping detected.



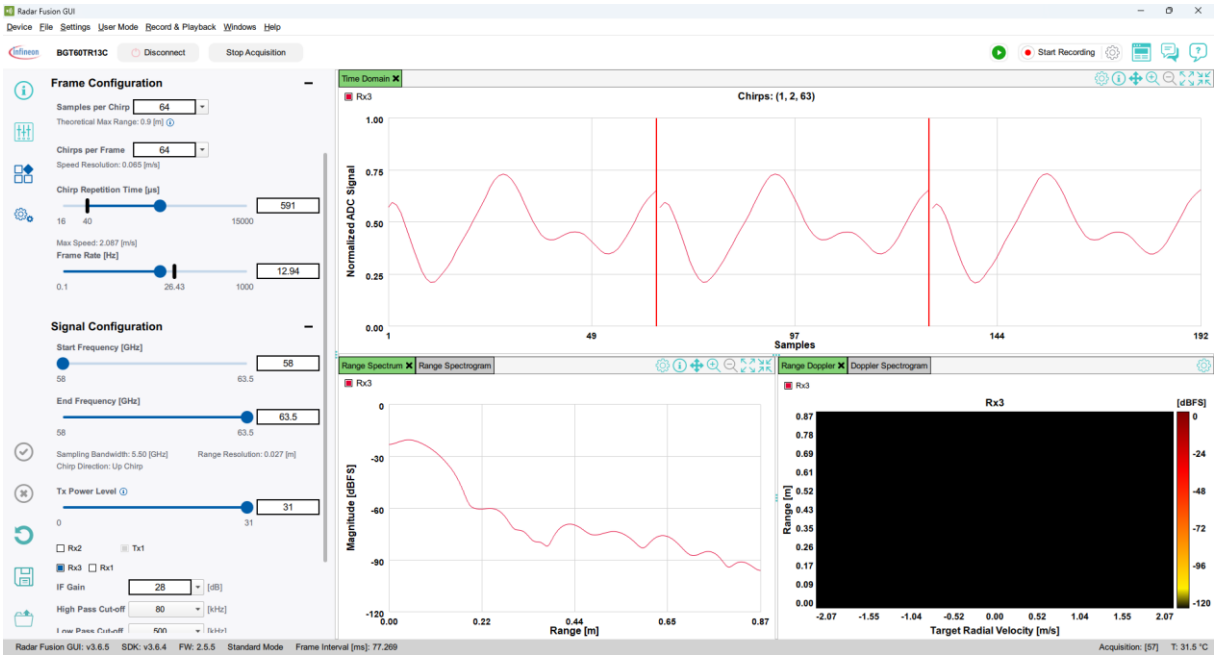
Water level = 220mm, gain = 28dB.

Result: correct.



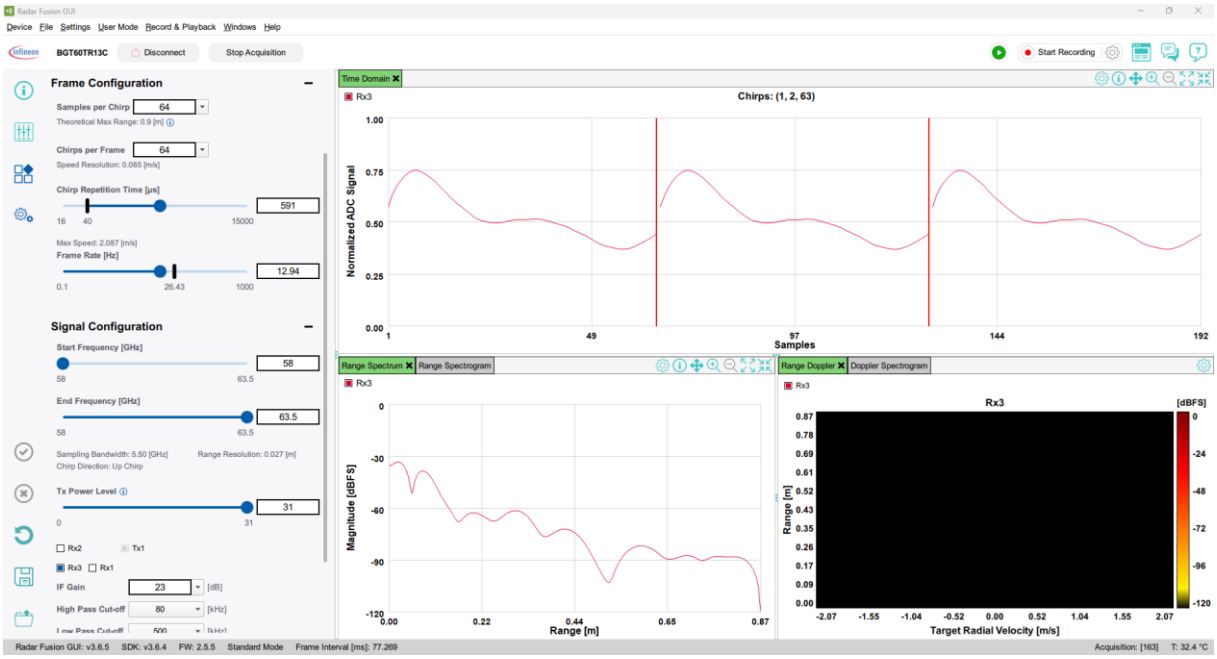
Water level = 240mm, gain = 33dB.

Result: saturated, clipping detected.



Water level = 240mm, gain = 28dB.

Result: correct.



Water level = 260mm, gain = 23dB.

Result: correct.

During our tests, we determined the minimum distance from the water level to the radar (290 – 260 = 30 mm) at which it produced accurate measurements under our test conditions. To achieve this result, we had to adjust the radar's IF Gain parameter.